

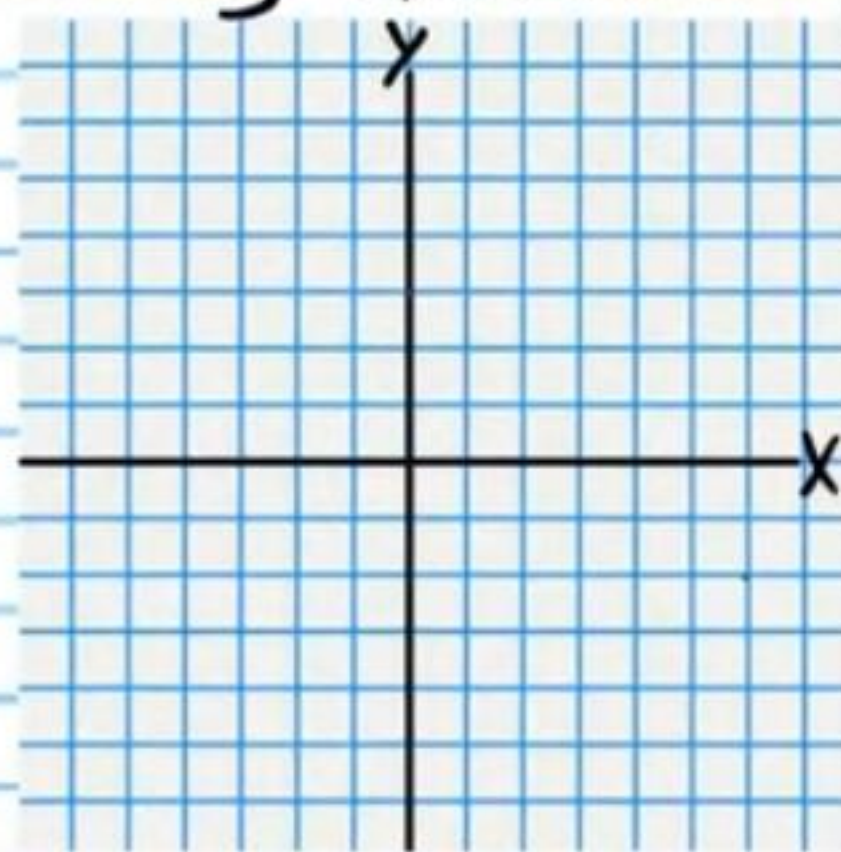
Intermediate Algebra Test I

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- II) Show all work.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

① Show the solutions of the following system of inequalities by graphing.

$$\begin{cases} y < -\frac{4}{3}x + 1 \\ 2x - 3y \geq 6 \end{cases}$$



② Determine if the following system has one solution, no solution, or infinite solutions. Prove algebraically.

$$\begin{cases} 5x - 2 = y \\ 15x - 3y = 6 \end{cases}$$

③ Find the algebraic solution and the graphical solution of the following inequality.

$$3 \geq -5|2x+4| + 23$$

④ Simplify. ~~Do not~~ assume that x is positive.

$$\sqrt[3]{x^{12}}$$

⑤ Simplify
 9^{101}

Write in $a+bi$ form.

⑥ $(8+3i) + (4-5i)$

⑦ $(2-6i)(1-7i)$

⑧ $\frac{i-2}{5i}$

⑨ Find the algebraic solution and the graphical solution of the following inequality.

$$|x| \leq 0$$

⑩ Find the discriminant of the following equation and use it to determine the number of and type of solutions. Write "two real solutions," "one real solution," or "two complex solutions."

$$0 = -2x^2 + 4x - 5$$

⑪ Simplify. Assume that x is positive.

$$\sqrt[3]{16x} \cdot \sqrt[3]{4x^9}$$

⑫ Simplify. Assume that x is positive.

$$\sqrt[4]{\frac{1}{4x^2}}$$

⑬ Solve.

$$\begin{cases} y = -2z \\ x + y + z = -6 \\ -x - y + z = 8 \end{cases}$$

⑭ Solve.

$$-8 > \frac{|\frac{x}{2} + 5|}{3} + 4$$

⑮ Rationalize the denominator.

$$\frac{\sqrt{2} - 2}{\sqrt{3} + 2}$$

①6 Solve.

$$\begin{cases} 5x + 2y - 3z = 4 \\ -3x - y + 2z = -2 \\ 2x + 4y - z = 5 \end{cases}$$

Solve problem #'s 17-20 by completing the square.

①7 $x^2 - 8x + 15 = 0$

①8 $-4x^2 + 16x + 180 = 0$

$$(19) x^2 + x + 2 = 0$$

$$(20) 2x^2 - 18x - 10 = 0$$

Solve problem #'s 21-22 using the quadratic formula.

$$(21) 2x^2 - x - 1 = 0$$

$$(22) -x^2 + 3x - 3 = 0$$

②③ Solve.

$$|-2x| + 1 = -5$$

②④ Simplify and rationalize the denominator.

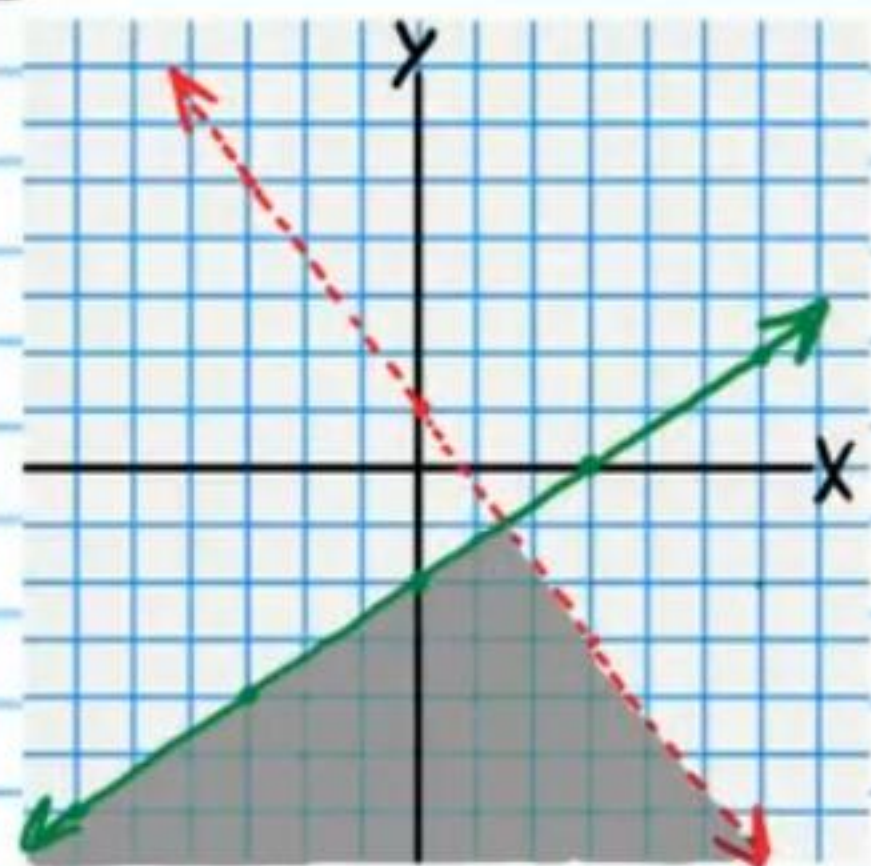
$$\sqrt{\frac{36}{5x}}$$

②⑤ Simplify and rationalize the denominator.

$$\sqrt{\frac{54}{125}}$$

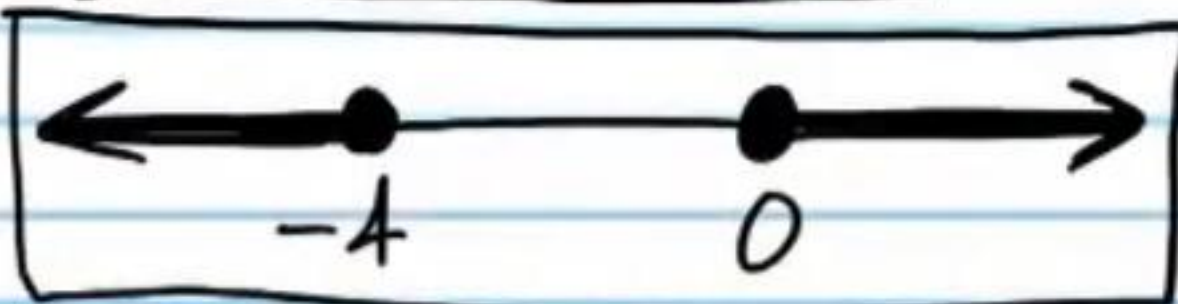
Answers

①



② Infinite solutions

③ $x \leq -4$ or $x \geq 0$



④ x^4 Absolute value symbols are okay but not necessary.

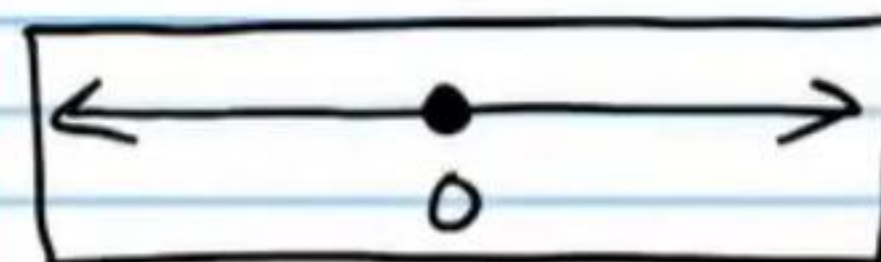
⑤ i

⑥ $12 - 2i$

⑦ $-40 - 20i$

⑧ $\frac{1}{5} + \frac{2}{5}i$

⑨ $x = 0$



⑩ $D = -24$

Two complex solutions

⑪ $4x^3 \sqrt[3]{x}$

⑫ $\frac{\sqrt[4]{4x^2}}{2x}$

⑬ $(-5, -2, 1)$

⑭ No solution

⑮ $-\sqrt{6} + 2\sqrt{2} + 2\sqrt{3} - 4$

⑯ $(1, 1, 1)$

⑰ $x = 5$ or $x = 3$

⑱ $x = 9$ or $x = -5$

⑲ $x = \frac{-1 \pm i\sqrt{7}}{2}$

$$(20) \quad x = \frac{9 \pm \sqrt{101}}{2}$$

$$(21) \quad x = 1 \text{ or } x = -\frac{1}{2}$$

$$(22) \quad x = \frac{3 \pm i\sqrt{3}}{2}$$

(23) No solution

$$(24) \quad \frac{6\sqrt{5}x}{5x}$$

$$(25) \quad \frac{3\sqrt{30}}{25}$$